1BM24CS315 WEEK-05

1. Implement Simulated Annealing algorithm for 8 queens.

   Output:

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========== Restart 1 ==========
Initial state: [1, 2, 3, 4, 5, 6, 7, 8]
Initial cost: 28

Initial grid:
Q . . . . . . .
. Q . . . . . .
. . Q . . . . .
. . . Q . . . .
. . . . Q . . .
. . . . . Q . .
. . . . . . Q .
. . . . . . . Q

Step 1
Current state: [1, 2, 3, 4, 5, 6, 7, 8]
Current cost: 28
Neighbor state: [1, 2, 3, 4, 5, 6, 2, 8]
Neighbor cost: 22
Delta E: -6
Temperature: 10.0000
Acceptance probability: 1.0000
Decision: ACCEPT

Grid after Step 1 (cost 22):
Q . . . . . . .
. Q . . . . Q .
. . Q . . . . .
. . . Q . . . .
. . . . Q . . .
. . . . . Q . .
. . . . . . . .
. . . . . . . Q
```